Pierre Mercuriali

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PierreMercuriali

Current

2020 - 2021 **PostDoc at TUDelft, Department of Cognitive Robotics**, *Delft*, Ontologies and formal reasoning to capture tool use among primates and ancient humans, the Netherlands.

Education

2016-2020 **PhD in Theoretical Computer Science. Directors : Miguel Couceiro et Romain Péchoux**, *LORIA (Laboratoire Lorrain de Recherche en Informatique et ses Applications)*, Nancy, Title : On normal form systems to efficiently represent multivariate functions over finite sets (defended December 2020).
keywords: median, clone theory, computational complexity, Boolean logic, knowledge representation, multi-valued logic, normal forms

2016 **Master's in Cognitive Science, Natural Language Processing, Computational linguistics**, *UFR-MI (Université de Lorraine)*, Nancy, France.

2014 **Two year's degree in English**, *LLCER (Université de Lorraine)*, Nancy, France.

2012 **Bachelor's in Mathematics**, *UFRSTMIA*, Nancy, France.

2009 **Baccalauréat S–SVT (Biology)**, *Lycée Raymond Poincaré*, Bar-le-Duc, France.

Summer schools

2017 **ESSLLI 2017**, *29th European Summer School in Logic, Language, and Information*, Université de Toulouse, France.

2018 **EJCIM 2018**, *Écoles Jeunes Chercheurs en Informatique Mathématique*, LORIA, Nancy. France

Experience

Teaching

- 2020-2021: Formal Logic (as part of RO47014: Knowledge Representation and Symbolic Reasoning course), TUDelft
- 2016-2018: C2i (Information and communications technology certificate), Université de Lorraine
- 2017-2018: Algorithms et Programming, Université de Lorraine

Conferences and workshops

- June 2021: CAA2021 - Computer Applications in Archaeology, Limassol, Cyprus (+ online)
- Avril 2018: DICE 2018 - Developments in Implicit Computational Complexity, Thessaloniki, Greece
- Octobre 2017: LFA 2017 - 26th Rencontres Francophones sur la Logique Floue et ses Applications, Amiens, France

Publications

[CLM+18] Miguel Couceiro, Erkko Lehtonen, Pierre Mercuriali, Romain Péchoux, and Mathias Soeken. Normal form systems generated by single connectives have mutually equivalent efficiency. In *DICE 2018 - Developments in Implicit Computational Complexity*, Thessaloniki, Greece, April 2018.

[CLMP20] Miguel Couceiro, Erkko Lehtonen, Pierre Mercuriali, and Romain Péchoux. On the efficiency of normal form systems for representing Boolean functions. *Theor. Comput. Sci.*, 813:341–361, 2020.

[CMP17] Miguel Couceiro, Pierre Mercuriali, and Romain Péchoux. On the efficiency of normal form systems of Boolean functions. In *LFA 2017 - 26èmes Rencontres Francophones sur la Logique Floue et ses Applications*, pages 1–8, Amiens, France, October 2017.

[CMPS17] Miguel Couceiro, Pierre Mercuriali, Romain Péchoux, and Abdallah Saffidine. Median based calculus for lattice polynomials and monotone Boolean functions. In *47th IEEE International Symposium on Multiple-Valued Logic, ISMVL 2017, Novi Sad, Serbia, May 22-24, 2017*, pages 37–42. IEEE Computer Society, 2017.

[CMPS19] Miguel Couceiro, Pierre Mercuriali, Romain Péchoux, and Abdallah Saffidine. On the complexity of minimizing median normal forms of monotone Boolean functions and lattice polynomials. *J. Multiple Valued Log. Soft Comput.*, 33(3):197–218, 2019.

Computer skills

- Office: LibreOffice, LaTeX
- Programming: Python (NLTK, PIL, numpy), C, Java, Prolog, web development (HTML+CSS, javascript), Linux
- Digital painting: Krita, Photoshop

Languages

French	Native
English	C2 level; daily practice, professional use
Italian	B1/B2 level; regular practice
Ancient Greek, Dutch, Latin, Polish	Bases

Academic interests

- Formal logic, mathematics
- Computational linguistics, natural language processing
- Cognitive science, artificial intelligence, knowledge representation
- Education

Misc. Interests

- Science-fiction, French literature, ancient philosophy, linguistics and languages
- Ancient history, history of science, diachronic linguistics
- Production and musical practice, traditional music (especially Celtic)
- Academic drawing, illustration, painting (digital), visual mathematics
- Gardening